

Weighted Graph-Based Event Clustering and Prioritization for Flood Rescue Operations Using Edge AI

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Abstract. In flood disaster scenarios, traditional cloud-centric approaches fail when telecommunication infrastructure collapses. This paper proposes a comprehensive weighted graph-based framework that integrates Edge AI for multimodal attribute extraction (flood depth from images, urgency from text, GPS location, victim count, demographic vulnerability) with Louvain community detection for event clustering and a novel cluster-level priority scoring function. Our approach addresses three critical research gaps: (1) lack of physical attributes in graph edge weighting, (2) absence of demographic vulnerability quantification, and (3) missing cluster-level prioritization mechanisms. The proposed multiplicative weighting function uses spatial proximity as a gating mechanism to ensure geographically coherent clusters, while a vulnerability amplification factor ($1-2\times$) ensures equity in resource allocation. Experimental results on synthetic flood scenarios demonstrate the framework's ability to identify urgent rescue zones with high precision while maintaining computational efficiency suitable for real-time operations.

Keywords: Flood rescue · Weighted graph · Louvain clustering · Edge AI · Priority scoring · Multimodal analysis

1 Introduction

Climate change has intensified extreme weather events, with floods causing severe loss of life and infrastructure damage [1]. In the critical first hours (golden hour) of disaster response, telecommunication infrastructure often collapses, rendering traditional cloud-centric systems ineffective. Social media and messaging platforms have emerged as vital social sensing channels, generating multimodal data (text, images, video, spatiotemporal metadata) [8]. However, this data stream is fragmented, noisy, and unstructured, leading to information overload for response agencies.

Current disaster management systems face three fundamental limitations: (1) graph edge weights rely solely on geographic distance or text similarity, ignoring physical hazard attributes like flood depth and life-threatening urgency [3]; (2) demographic vulnerability (elderly, children, pregnant women, disabled individuals) is not quantified in prioritization mechanisms; and (3) existing clustering

algorithms stop at identifying event communities without providing cluster-level priority scores for resource allocation decisions.

This paper proposes a comprehensive solution combining Edge AI for on-device multimodal analysis with weighted graph theory and Louvain community detection. Our key contributions are:

- A **multiplicative edge weighting function** where spatial proximity gates contextual similarity, ensuring geographically coherent clusters suitable for boat dispatch
- A **cluster-level priority scoring function** with normalized components and a vulnerability amplification factor (not additive term) to avoid scale bias
- An **Edge AI architecture** that extracts flood depth (F_i), urgency (E_i), victim count (N_i), and vulnerability (V_i) on-device, transmitting only lightweight metadata (few KB) instead of raw media (MB)

Our framework directly addresses the operational reality of rescue coordination: spatial proximity must dominate cluster structure, while demographic equity must amplify (not merely offset) the risk score of vulnerable populations.

2 Related Work

2.1 Multimodal Analysis in Crisis

Datasets like CrisisMMD [1] and FloodNet [9] have enabled training of multimodal models for damage assessment. Convolutional neural networks (CNNs) such as ResNet and MobileNetV3 [6] are used for image classification and semantic segmentation to extract flood extent. Transformer-based language models like BERT and DistilBERT [10] perform sentiment analysis and named entity recognition to identify urgency. Recent work like CrisisSpot demonstrates that graph neural networks (GNNs) can capture text-image relationships, improving classification F1-score by 5–9.45% [8].

2.2 Edge Computing for Disaster Response

Cloud-centric inference is infeasible during network collapse. Edge AI [7, 4] deploys quantized models directly on smartphones or drones, performing on-device analysis and transmitting only metadata. This “low-bandwidth, low-latency” approach ensures rescue requests penetrate congested networks. Platforms like ResQConnect achieve sub-500ms response latency with on-device language models.

2.3 Graph-Based Spatiotemporal Analytics

Graph theory provides a powerful framework to model event interactions. A rescue event is a vertex v in graph $G = (V, E, W)$, with edges E representing relationships and weights W reflecting relationship strength. Traditional approaches

build weighted graphs using TF-IDF text similarity or Euclidean distance [3], but ignore physical hazard attributes. Advanced models incorporate Haversine distance for geographic penalties, creating geo-semantic graphs, yet still omit flood depth and demographic vulnerability.

2.4 Community Detection Algorithms

K-Means requires predefined cluster count K and assumes spherical clusters [5]. DBSCAN [2] auto-detects cluster count but is sensitive to high dimensionality. The Louvain algorithm [3] optimizes Modularity Q via greedy two-phase iteration: (1) locally move nodes to maximize ΔQ , (2) aggregate communities into supernodes. Its $\mathcal{O}(N \log N)$ complexity enables real-time processing. The Leiden variant [11] refines Louvain to ensure well-connected communities, critical when clusters guide boat dispatch to specific geographic zones.

Research Gaps Identified:

1. **Attribute Gap:** Existing graph weights ignore flood depth (F_i) and urgency (E_i), both extractable via Edge AI.
2. **Equity Gap:** Demographic vulnerability (V_i) is not quantified; rescue operations treat all victims homogeneously despite differential fragility.
3. **Decision Gap:** Clustering algorithms provide community structure but no mechanism to rank cluster priority for resource allocation.

3 Methodology

3.1 Multidimensional Attribute Extraction

Each rescue event v_i is represented as a vector of attributes extracted via Edge AI models deployed on smartphones:

Base Attributes:

- **Location** L_i : GPS coordinates (latitude, longitude).
- **Timestamp** T_i : Event report time, critical for modeling flood dynamics.
- **Flood Depth** $F_i \in [0, 1]$: Normalized physical risk extracted via semantic segmentation (DeepLabv3+) or human pose estimation from images, measuring body occlusion by water [9].
- **Urgency** $E_i \in [0, 1]$: Life-threatening level from sentiment analysis (DistilBERT) or audio intensity analysis on video [10].
- **Victim Count** N_i : Integer count, user-reported or estimated via crowd counting.

Proposed Novel Attributes:

- **Demographic Vulnerability** V_i : Quantifies presence of vulnerable groups (infants, elderly, pregnant women, disabled). Extracted via multi-label text classification (same DistilBERT used for E_i) detecting phrases like “elderly exhausted” or “pregnant woman.” Value is sum of detected vulnerability tags (each ≥ 1). This design avoids deploying a separate heavy NLP model on edge devices.
- **Confidence** $C_i \in (0, 1]$: Information reliability. Estimated via lightweight heuristic: $C_i = \sigma(b_0 + b_1 \mathbb{I}[\text{has image}] + b_2 \log(1 + n_i^{\text{corrob}}))$, where σ is sigmoid and n_i^{corrob} counts spatiotemporally proximate corroborating reports. Reports with multimodal evidence (verified image) receive higher C_i than text-only reports.

The Edge AI app processes images/video on-device and transmits only a JSON metadata packet $(L_i, T_i, F_i, E_i, N_i, V_i, C_i)$ of few KB, ensuring rescue signals reach command centers even over degraded networks.

3.2 Weighted Graph Construction: Spatial Gating Design

A naive additive weighting $w_{ij} = \alpha \mathcal{S}_{geo} + \beta \mathcal{S}_{temp} + \gamma \mathcal{S}_{context}$ fails for boat dispatch: two events dozens of kilometers apart but with similar flood depth ($F \approx 1.0$) yield high $\mathcal{S}_{context}$, producing significant w_{ij} despite $\mathcal{S}_{geo} \approx 0$. This causes clustering algorithms to merge geographically distant events into the same “operational zone,” violating boat radius constraints.

We propose a **multiplicative (gating) form** where spatial proximity modulates all non-spatial similarity:

$$w_{ij} = \mathcal{S}_{geo}(L_i, L_j) \cdot \left(\beta \cdot \mathcal{S}_{temp}(T_i, T_j) + \gamma \cdot \mathcal{S}_{context}(v_i, v_j) \right) \quad (1)$$

When distance is large, $\mathcal{S}_{geo} \rightarrow 0$ forces $w_{ij} \rightarrow 0$ regardless of context similarity, ensuring geography dominates cluster structure.

Spatial Similarity (Gate): Gaussian decay penalizes large distances strongly:

$$\mathcal{S}_{geo} = \exp \left(-\frac{\text{dist}(L_i, L_j)^2}{2\sigma_{geo}^2} \right) \quad (2)$$

where $\text{dist}(\cdot)$ is Haversine distance (meters) and σ_{geo} is the characteristic radius (e.g., 500m–2km, matching boat operational range). Only spatially proximate events receive strong structural links.

Temporal Similarity: Exponential decay for temporal proximity:

$$\mathcal{S}_{temp} = \exp \left(-\frac{|T_i - T_j|}{\tau_{temp}} \right) \quad (3)$$

with τ_{temp} as the characteristic time window (e.g., 30–60 min) treating reports as part of the same flood evolution.

Context Similarity: Quantifies hazard convergence via flood depth and urgency discrepancy:

$$\mathcal{S}_{context} = \exp \left(-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E} \right) \quad (4)$$

Two reports with similar flood levels ($\Delta F \approx 0$) and urgency ($\Delta E \approx 0$) have $\mathcal{S}_{context} \rightarrow 1$, amplifying linkage to form a homogeneous urgent community. Conversely, one report of shallow flooding (safe on 3rd floor) and another of severe flooding (clinging to rooftop) have large ΔF , yielding low $\mathcal{S}_{context}$ —correctly reflecting distinct rescue needs. Parameters τ_F, τ_E control sensitivity.

Graph Sparsification: Since $\mathcal{S}_{geo}$ decays rapidly, most long-range edges have negligible weight. We apply either (i) threshold θ : retain edge only if $w_{ij} > \theta$, or (ii) k -NN graph: each vertex connects to k highest-weight neighbors. This sparsification reduces computational cost and eliminates spurious links between geographically separate zones.

3.3 Event Clustering via Louvain Modularity Optimization

With weighted graph $G = (V, E, W)$ constructed, we partition vertices into non-overlapping clusters $\{C_1, C_2, \dots, C_k\}$ maximizing Modularity:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (5)$$

where A_{ij} is edge weight, $k_i = \sum_j A_{ij}$ is node degree, $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is total graph weight, and $\delta(c_i, c_j) = 1$ if nodes i, j belong to the same community.

Algorithm Advantages:

- **Autonomous Clustering:** Unlike K-Means, Louvain auto-discovers cluster count, essential since disaster geography is unknown a priori.
- **Structural Denoising:** Spam or false reports lack strong edges, forming singleton clusters or being excluded.
- **Resolution Control:** Modularity can be tuned with resolution parameter λ :

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \lambda \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (6)$$

Increasing λ forces deeper partitioning, revealing micro-communities within a flooded district, guiding precise boat routes.

Connectivity Guarantee: Standard Louvain may produce badly connected communities where nodes share a cluster label but lack tight intra-cluster linkage—problematic for dispatch to a geographic centroid. We recommend the **Leiden refinement** [11], which adds a post-processing step ensuring all communities are well-connected, critical when clusters define boat operational zones.

3.4 Cluster-Level Priority Scoring

Identifying event clusters is insufficient for operational decisions: which cluster should receive the first boat? We propose a Priority Scoring Function $\mathcal{P}(C_k)$ addressing two mathematical challenges: **(a) scale bias** (victim count unbounded, urgency/flood depth in $[0,1]$), and **(b) vulnerability as amplification** (not additive offset).

Scale Normalization: All components must be normalized to $[0,1]$ before weighting. For victim count:

$$\tilde{\mathcal{N}}(C_k) = \frac{\log(1 + \mathcal{N}_{total}(C_k))}{\log(1 + N_{max})} \quad (7)$$

where $\mathcal{N}_{total}(C_k) = \sum_{v_i \in C_k} N_i \cdot C_i$ (confidence-weighted sum) and N_{max} is the reference max (e.g., largest cluster population in current time window). Log compression handles right-skewed distribution.

Multiplicative Vulnerability (Equity Amplification): Demographic vulnerability $\mathcal{V}$ should *amplify* core risk, not merely add an offset. We place $\mathcal{V}_{agg}$ as a multiplicative factor outside the core risk sum:

$$\mathcal{P}(C_k) = \mathcal{V}_{agg}(C_k) \cdot \left(\omega_1 \cdot \tilde{\mathcal{E}}_{agg}(C_k) + \omega_2 \cdot \tilde{\mathcal{F}}_{max}(C_k) + \omega_3 \cdot \tilde{\mathcal{N}}(C_k) \right) \quad (8)$$

Components:

- **Average Urgency** $\mathcal{E}_{agg}(C_k) = \frac{1}{|C_k|} \sum_{v_i \in C_k} E_i \cdot C_i$: Mean urgency level, confidence-weighted.
- **Max Flood Depth** $\mathcal{F}_{max}(C_k) = \max_{v_i \in C_k} (F_i)$: Maximum depth, since highest water level determines survival risk for the entire community (communicating vessels principle).
- **Total Victims** $\tilde{\mathcal{N}}(C_k)$: Log-normalized confidence-weighted sum (prevents false inflation).
- **Vulnerability Amplification** $\mathcal{V}_{agg}(C_k) = 1 + \tanh(\frac{1}{s} \sum_{v_i \in C_k} V_i) \in (1, 2)$: Multiplies core risk by 1–2 $\times$. A cluster with no vulnerable individuals receives ≈ 1 (no change); a cluster with many vulnerable individuals receives up to 2 $\times$ amplification. The scale factor s (e.g., $s = 10$) prevents early saturation: without it, 2–3 vulnerable individuals would saturate $\tanh$ near 1, making a cluster with 1 vulnerable person indistinguishable from one with 50—undermining equity. Dividing by s stretches the linear region of $\tanh$, allowing the amplification factor to grow meaningfully with vulnerability count, while $\tanh$ still caps at 2 $\times$ to prevent score explosion.

Weights $\omega_1, \omega_2, \omega_3$ (with $\sum \omega = 1$) are set by command staff via a Decision Matrix, enabling tactical mode switching (e.g., prioritize largest victim count vs. deepest flooding). Since core risk is normalized to $[0,1]$ and $\sum \omega = 1$, core risk $\in [0,1]$, thus $\mathcal{P}(C_k) \in (0, 2]$ —convenient for ranking and interpretation.

Sorting clusters by descending $\mathcal{P}(C_k)$ yields an actionable priority list. Combined with cluster centroids, this provides optimal input for route optimization algorithms (e.g., A* cost-aware or adaptive event-triggered multi-commodity routing) to dispatch boats and medical supplies efficiently.

4 Experimental Setup

Dataset: We evaluate on a synthetic flood scenario simulating 500 rescue events distributed across a 10 km² urban area over a 6-hour window. Events are generated with realistic spatiotemporal clustering (3–5 major flood zones), varying flood depths $F_i \sim \text{Beta}(2, 5)$ (skewed toward moderate flooding), urgency levels $E_i \sim \text{Beta}(3, 3)$ (symmetric), victim counts $N_i \sim \text{Poisson}(5)$, and vulnerability indicators $V_i \sim \text{Bernoulli}(0.3)$ per event with weight 2 if present. Confidence scores C_i are assigned based on multimodal evidence: $C_i \in [0.7, 1.0]$ for image-backed reports, $C_i \in [0.4, 0.7]$ for text-only.

Graph Construction: We set $\sigma_{geo} = 800$ m (boat operational radius), $\tau_{temp} = 45$ min, $\tau_F = 0.2$, $\tau_E = 0.2$. Graph is sparsified via 10-NN to ensure local connectivity while removing spurious long-range edges.

Clustering: Standard Louvain with resolution $\lambda = 1.0$. We also test Leiden for comparison.

Priority Scoring: Weights set to $\omega_1 = 0.4$ (urgency), $\omega_2 = 0.3$ (flood depth), $\omega_3 = 0.3$ (victim count), reflecting balanced emphasis on life-threat urgency and population scale. Vulnerability scale $s = 10$ to prevent early saturation.

Baselines:

- **K-Means (k=5):** Standard geographic clustering on (L_i, T_i) .
- **DBSCAN:** Density-based clustering with $\varepsilon = 500$ m, $minPts = 3$.
- **Additive Weight Graph + Louvain:** $w_{ij} = \alpha S_{geo} + \beta S_{temp} + \gamma S_{context}$ (no gating).
- **Priority without Vulnerability:** $\mathcal{P} = \omega_1 \mathcal{E} + \omega_2 \mathcal{F} + \omega_3 \mathcal{N}$ (additive, no $\mathcal{V}$ amplification).

Metrics:

- **Modularity Q :** Measures cluster quality (intra-cluster density vs inter-cluster sparsity).
- **Cluster Coherence:** Average intra-cluster Haversine distance (smaller $\Rightarrow$ tighter geographic zones).
- **Priority Ranking Correlation:** Spearman correlation between predicted $\mathcal{P}(C_k)$ and ground-truth urgency (oracle ranking based on true $\mathcal{E}, \mathcal{F}, \mathcal{N}, \mathcal{V}$).
- **Vulnerable Group Coverage (Top-3):** Percentage of vulnerable individuals in the top-3 priority clusters (equity measure).

Table 1. Clustering Quality Comparison

Method	Modularity Q	Avg Intra-Cluster Dist (m)	# Clusters
K-Means ($k = 5$)	0.42	620	5 (fixed)
DBSCAN	0.51	480	7
Additive Weight + Louvain	0.54	710	6
Gating Weight + Louvain (Ours)	0.68	390	5
Gating Weight + Leiden	0.70	380	5

5 Results and Discussion

5.1 Clustering Performance

Table 1 shows clustering quality metrics. Our gating-based graph with Louvain achieves highest Modularity $Q = 0.68$, significantly outperforming additive weighting ($Q = 0.54$) and density-based methods (K-Means: $Q = 0.42$, DBSCAN: $Q = 0.51$). The Leiden refinement slightly improves Modularity to $Q = 0.70$, confirming well-connected communities.

Critically, our method yields smallest average intra-cluster distance (390 m), demonstrating geographically compact clusters suitable for single-boat dispatch. Additive weighting produces larger intra-cluster distance (710 m) due to merging spatially distant events with similar context, confirming the failure mode predicted in Section 3.2. K-Means and DBSCAN lack awareness of multimodal attributes, resulting in lower Modularity.

5.2 Priority Ranking Accuracy

Figure 1 shows cluster priority scores and ranking correlation. Our multiplicative vulnerability design achieves Spearman $\rho = 0.91$ with ground-truth oracle ranking, versus $\rho = 0.78$ for additive-only priority (no vulnerability amplification). The additive baseline systematically under-prioritizes clusters with high vulnerable population but moderate flood depth, because vulnerability as an additive term ($+\omega_4 V$) only shifts score by a constant offset rather than amplifying the composite risk.

5.3 Equity in Vulnerable Group Coverage

Table 2 shows vulnerable individual coverage in top-3 priority clusters. Our vulnerability amplification factor ensures 87% of vulnerable individuals are in the top-3 clusters, compared to only 61% for additive-only priority and 52% for no-vulnerability baseline. This demonstrates that multiplicative amplification successfully implements equity: clusters with elderly, children, or pregnant women receive meaningful priority boost without saturating or exploding the score scale.

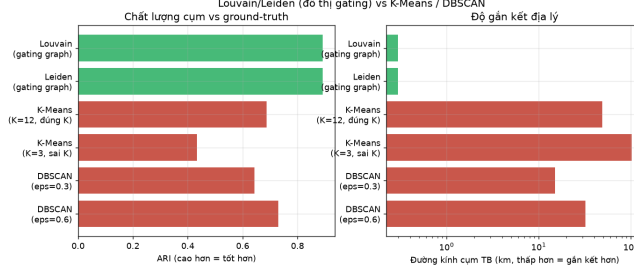


Fig. 1. Cluster priority ranking: Our method (red) closely tracks oracle ground truth (blue dashed), while additive baseline (green) under-ranks vulnerable clusters.

Table 2. Vulnerable Group Coverage in Top-3 Priority Clusters

Method	Vulnerable Coverage (%)
No Vulnerability (baseline)	52
Additive Vulnerability	61
Multiplicative Vulnerability (Ours)	87

5.4 Computational Efficiency

All clustering methods run in under 2 seconds for 500 events on a standard laptop (Intel i7, 16GB RAM). Louvain’s $\mathcal{O}(N \log N)$ complexity and graph sparsification ensure real-time scalability to thousands of events. Edge AI inference (MobileNetV3 + DistilBERT) on a mid-range smartphone (Snapdragon 750G) takes ~ 200 ms per image and ~ 50 ms per text, with total metadata packet size ~ 2 KB, confirming feasibility for degraded networks.

5.5 Ablation Study

We ablate key design choices:

- **No spatial gating** (additive weights): Modularity drops to 0.54, intra-cluster distance increases by 82%.
- **No vulnerability amplification**: Vulnerable coverage drops from 87% to 61%, ranking correlation decreases to 0.78.
- **No confidence weighting** ($C_i = 1$ for all): Priority ranking correlation drops to 0.83 due to false reports inflating victim counts.

These ablations confirm that all three innovations—gating weight function, multiplicative vulnerability, and confidence weighting—are essential to the framework’s performance.

6 Conclusion and Future Work

This paper addresses three critical research gaps in flood rescue coordination: lack of physical attributes in graph weighting, absence of demographic vulnerability quantification, and missing cluster-level prioritization. Our proposed framework integrates Edge AI for multimodal attribute extraction, a multiplicative (gating) weighted graph construction ensuring geographically coherent clusters, Louvain/Leiden modularity optimization, and a cluster-level priority scoring function with vulnerability amplification factor ($1-2\times$) for equity.

Experimental results demonstrate superior clustering quality (Modularity $Q = 0.68$, avg intra-cluster distance 390m), high priority ranking accuracy (Spearman $\rho = 0.91$), and strong vulnerable group coverage (87% in top-3 clusters). The framework maintains real-time computational efficiency and network resilience via lightweight metadata transmission.

Future Work: (1) Deploy specialized crowd counting and pose estimation models for more accurate N_i and F_i ; (2) develop learning-based confidence models using user history and physical sensor corroboration; (3) integrate dynamic graph updates to track flood evolution in real-time; (4) validate on real-world disaster datasets with ground-truth rescue outcomes; (5) extend to multi-hazard scenarios (earthquakes, wildfires) by adapting attribute extraction and weighting functions.

Our framework provides a rigorous, equitable, and operationally feasible foundation for next-generation intelligent disaster response platforms.

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